

LECTURE 1

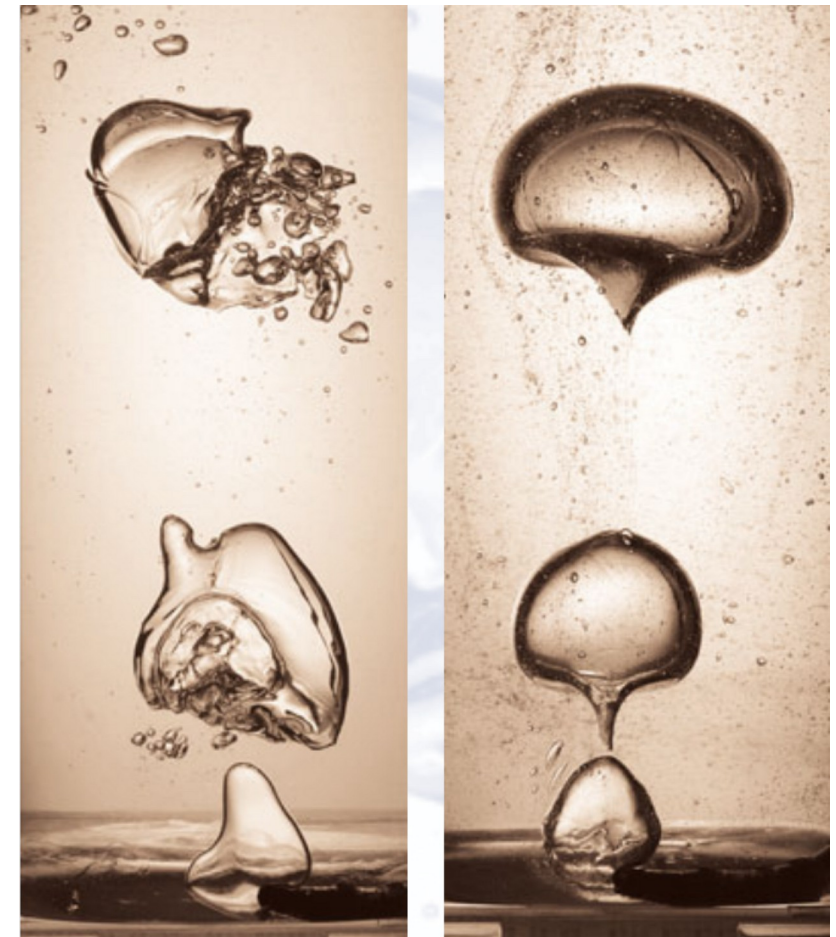
Introduction: Fluid Properties, Dimensions and Units

Reference: Munson et al., Fundamentals of Fluid Mechanics, Chapter 1
Wladimir E. Banda-Barragán

What is a fluid?

Matter is normally classified as solid or fluid, and fluids as liquids and gases. The difference is in the spacing of the molecules and in the forces between them.

- **Solids:** closely spaced molecules, strong cohesive forces, they hold their shape
- **Liquids:** molecules farther apart, weaker forces, easily deformed and hard to compress
- **Gases:** greater spacing still, negligible forces, compressible, they fill any container



The nature of air bubbles rising in a liquid is a function of fluid properties such as density, viscosity, and surface tension. (Left: air in oil; right: air in soap.)

How do we describe flow?

We require physical variables:

$$\ell \doteq L, \quad v \doteq LT^{-1}, \quad p \doteq FL^{-2} = ML^{-1}T^{-2}$$

- **Size ℓ :** the size of the body, or of the region the fluid occupies. From nanoscale devices, 10^{-8} m, to the diameter of Jupiter's Great Red Spot, 10^8 m — sixteen decades
- **Speed v :** how fast the fluid moves. From continental drift and glacier flow, through a river at about 1 m/s and the speed of sound in air, 343 m/s, to a meteor entering the atmosphere, above 10^4 m/s — twelve decades
- **Pressure p :** the normal force per unit area. From the faintest audible sound, 2×10^{-5} Pa, through the standard atmosphere, 101.3 kPa, a scuba tank at 20 MPa and the Mariana Trench at 110 MPa, to water-jet cutting near 400 MPa — 10 dex.

No single value of any one of them decides how a fluid behaves. What decides it are ratios: inertia against viscosity, speed against the speed of sound.

It is common to use dimensionless quantities.

The massive range of flow parameters

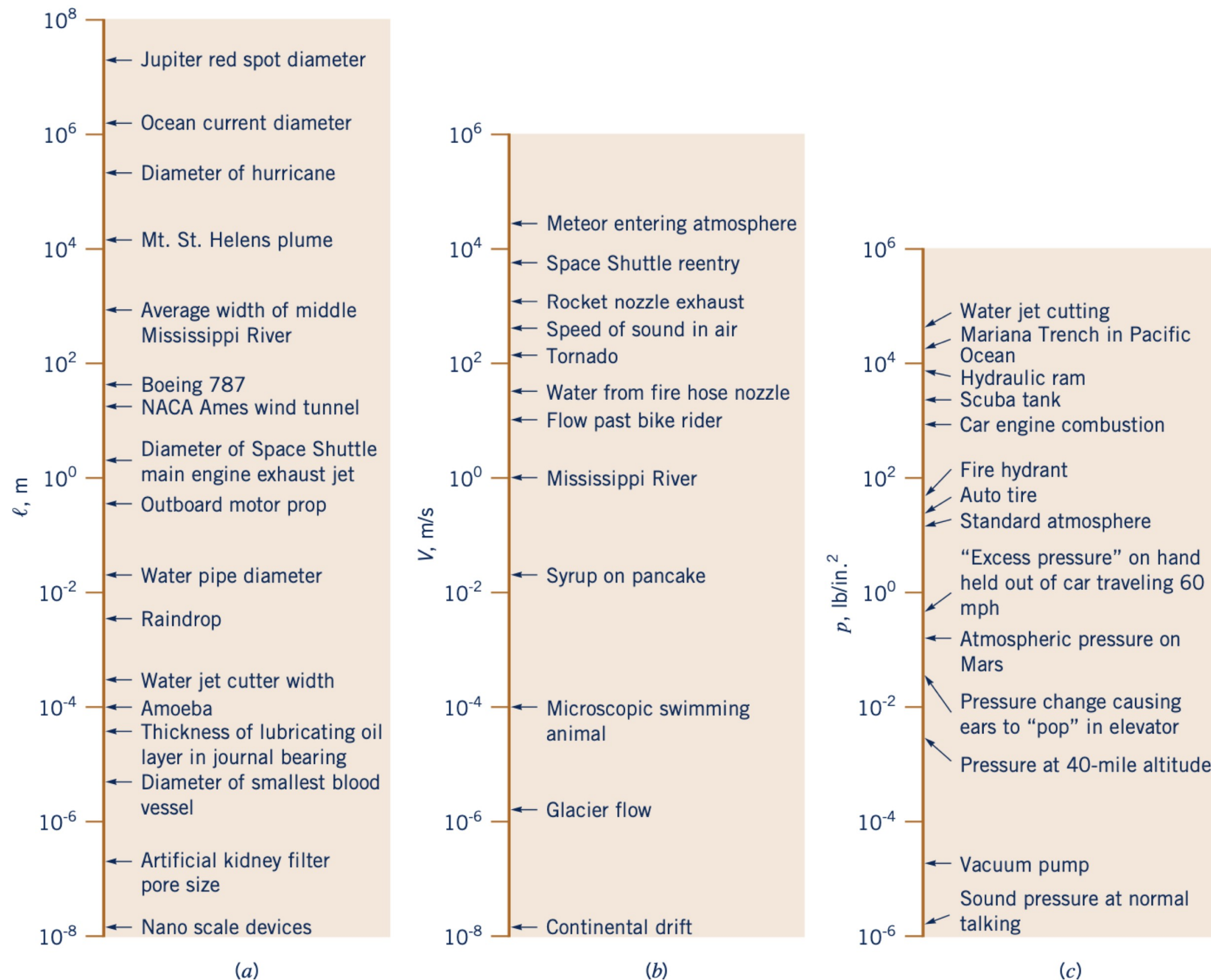
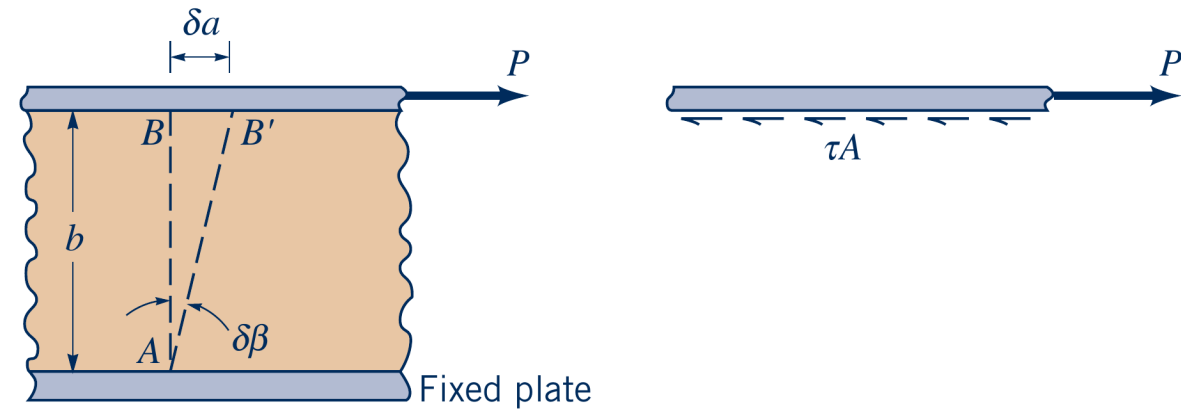


Figure 1.1 Characteristic values of some fluid flow parameters for a variety of flows: (a) object size, (b) fluid speed, (c) fluid pressure.

Molecular structure and deformation

- **Solid:** molecules packed at about one molecular diameter, held in place by strong cohesive forces. Each one vibrates about a fixed position in the lattice.
- **Liquid:** spacing slightly larger, forces weaker. Molecules slide past one another but stay in contact, so the volume is kept and the shape is not.
- **Gas:** spacing about ten molecular diameters, forces negligible. Molecules move freely and fill whatever contains them

$$d_{\text{H}_2\text{O}} \approx 0.3 \text{ nm}, \quad d_{\text{air}} \approx 3 \text{ nm}$$



A solid between two parallel plates, and the forces on the upper plate.

Apply a shearing stress. In a solid the bonds stretch, the material deforms through a small angle $\delta\beta$ and stops. In a fluid nothing holds the molecules in place, so the deformation never stops.

$$\text{solid: } \tau \propto \delta\beta$$

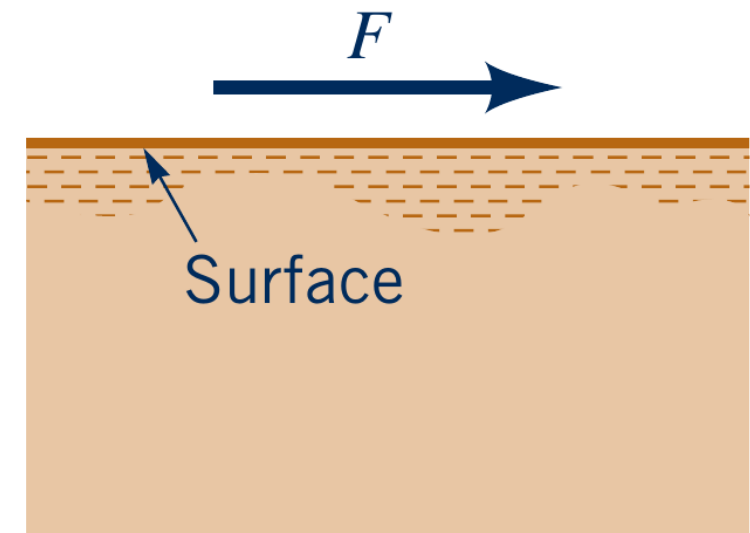
$$\text{fluid: } \tau \propto \frac{d\beta}{dt}$$

The definition of a fluid

A shearing stress is the force per unit area created whenever a tangential force acts on a surface.

A solid under shear deforms once, usually by a small amount, and then stops. A fluid keeps deforming as long as the stress is applied.

Borderline materials exist: slurries, tar, putty and toothpaste flow only above a critical stress. **Their study is rheology.**



A fluid is a substance that deforms continuously when acted on by a shearing stress of any magnitude.

The continuum hypothesis

We do not track individual molecules or fluid particles, it is not practical.

Every property is an average over a volume that is small compared with the system but large compared with the molecular spacing, and the fluid is then treated as a continuum in which ρ , p and v vary continuously from point to point.

Is this a reasonable approximation?

Generally, yes.

- A gas holds of order 10^{18} molecules per cubic millimetre; a liquid, about a thousand times more.
- The molecular spacing is a few nanometres, far below any length scale of the flow.
- The assumption fails in rarefied gases: at very high altitude the mean free path is no longer negligible.

Analysis of fluid behaviour

The study of fluid mechanics is done **qualitatively** (nature, type of fluid) and **quantitatively** (dynamical variables with units).

Fluid dynamics rests on the same laws used in the mechanics of solids:

- Newton's laws of motion
- Conservation of mass (continuity equation)
- First and second laws of thermodynamics (energy conservation and entropy increase)

Fluid dynamics splits into:

- **Fluid statics:** the fluid is at rest. Pressure distribution, hydrostatic forces, buoyancy
- **Fluid dynamics:** the fluid is in motion, and the properties defined below are all needed

The properties that come next — density, specific weight, specific gravity, the ideal gas law and viscosity — are the ones that make the difference between one flow and another.

Fluids: liquids and gases

LIQUIDS

- **Heavy:** $\rho \approx 10^3 \text{ kg/m}^3$. Water is 998 kg/m^3 at 20°C
- **Nearly incompressible:** a 1% change in volume needs 21.5 MPa
- **Molecules in contact:** mean spacing $d \approx 0.31 \text{ nm}$
- **Dense:** $n \approx 3.3 \times 10^{28}$ molecules per m^3
- Keeps its volume, takes the shape of the container, has a free surface
- Viscosity falls as the temperature rises

GASES

- **Light:** $\rho \approx 1 \text{ kg/m}^3$. Air is 1.20 kg/m^3 at 20°C
- **Highly compressible:** about 15,000 times more compressible than water
- **Molecules far apart:** mean spacing $d \approx 3.4 \text{ nm}$
- **Dilute:** $n \approx 2.5 \times 10^{25}$ molecules per m^3
- Fills the whole container, and has no free surface
- Viscosity rises as the temperature rises

$$\frac{\rho_{\text{liquid}}}{\rho_{\text{gas}}} \approx 830, \quad \frac{n_{\text{liquid}}}{n_{\text{gas}}} \approx 1300, \quad \frac{d_{\text{gas}}}{d_{\text{liquid}}} \approx 11$$

Both are fluids: neither resists a shearing stress. They differ in how tightly the molecules are packed.

Measures of fluid mass and weight

$$\rho = \frac{m}{V}$$

Density: mass per unit volume, in kg/m³.

$$\gamma = \rho g$$

Specific weight: weight per unit volume, in N/m³. It depends on the local gravity.

$$v_{\text{sp}} = \frac{V}{m} = \frac{1}{\rho}$$

Specific volume: volume per unit mass, in m³/kg. Used mainly in thermodynamics.

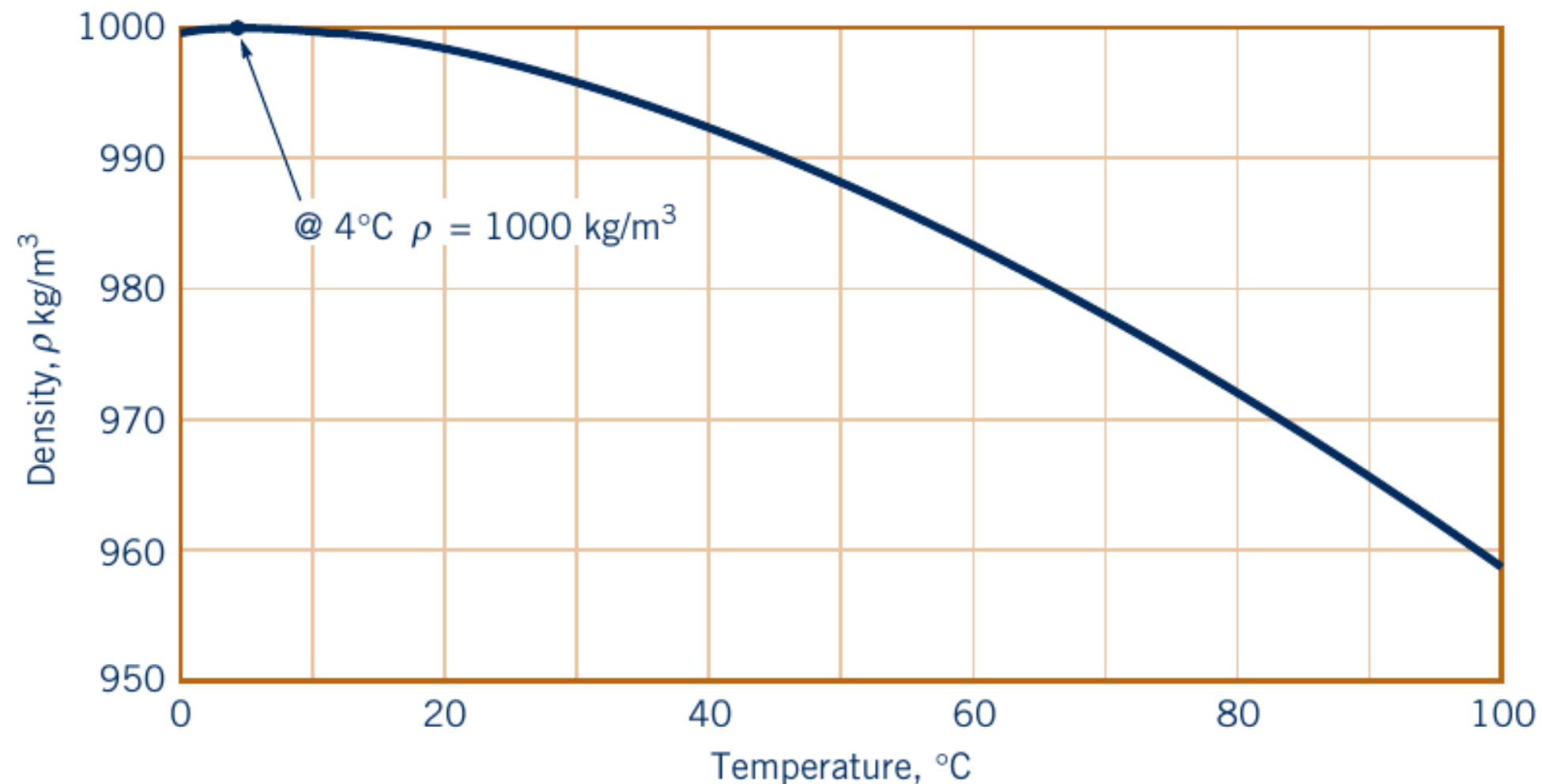
$$SG = \frac{\rho}{\rho_{\text{H}_2\text{O @ 4}^\circ\text{C}}}$$

Specific gravity: a ratio, so it is dimensionless.

Knowing any one of density, specific weight and specific gravity gives the other two.

Density of water

Density of water against temperature.



$$\rho_{\text{H}_2\text{O}} = 1000 \text{ kg/m}^3 \quad (4^\circ\text{C})$$

$$\gamma_{\text{H}_2\text{O}} = \rho g = 9.81 \text{ kN/m}^3$$

The density of a liquid is a weak function of temperature and almost independent of pressure. Between 4 °C and 100 °C the density of water falls by about 4%.

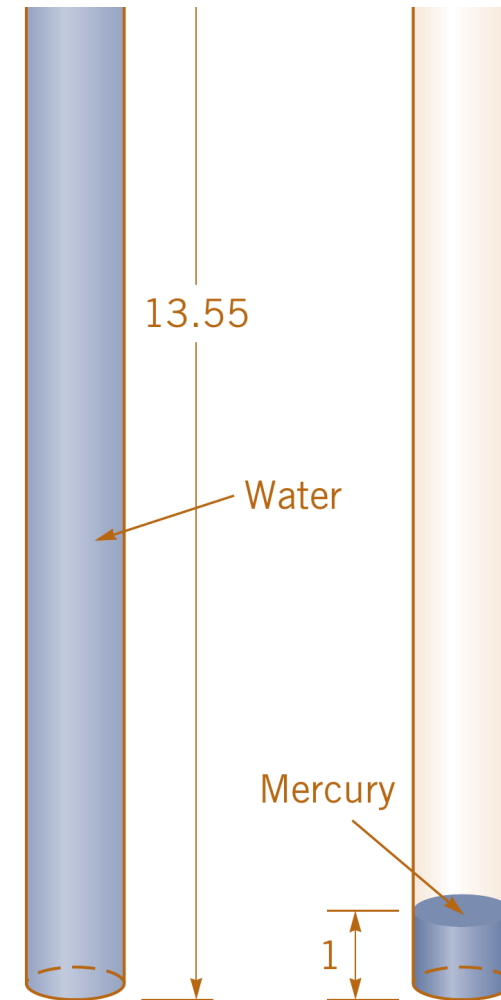
For gases both pressure and temperature matter, as the ideal gas law shows below.

Specific gravity

The specific gravity of mercury at 20 °C is 13.55: a column of mercury of a given height weighs 13.6 times as much as the same column of water.

$$\rho_{\text{Hg}} = (13.55)(1000) = 13.6 \times 10^3 \text{ kg/m}^3$$

Because it is a ratio of densities, SG does not depend on the system of units.



Properties of some common liquids

■ TABLE 1.6

Approximate Physical Properties of Some Common Liquids (SI Units)

Liquid	Temperature (°C)	Density, ρ (kg/m ³)	Specific Weight, γ (kN/m ³)	Dynamic Viscosity, μ (N · s/m ²)	Kinematic Viscosity, ν (m ² /s)	Surface Tension, ^a σ (N/m)	Vapor Pressure, p_v [N/m ² (abs)]	Bulk Modulus, ^b E_v (N/m ²)
Carbon tetrachloride	20	1,590	15.6	9.58 E − 4	6.03 E − 7	2.69 E − 2	1.3 E + 4	1.31 E + 9
Ethyl alcohol	20	789	7.74	1.19 E − 3	1.51 E − 6	2.28 E − 2	5.9 E + 3	1.06 E + 9
Gasoline ^c	15.6	680	6.67	3.1 E − 4	4.6 E − 7	2.2 E − 2	5.5 E + 4	1.3 E + 9
Glycerin	20	1,260	12.4	1.50 E + 0	1.19 E − 3	6.33 E − 2	1.4 E − 2	4.52 E + 9
Mercury	20	13,600	133	1.57 E − 3	1.15 E − 7	4.66 E − 1	1.6 E − 1	2.85 E + 10
SAE 30 oil ^c	15.6	912	8.95	3.8 E − 1	4.2 E − 4	3.6 E − 2	—	1.5 E + 9
Seawater	15.6	1,030	10.1	1.20 E − 3	1.17 E − 6	7.34 E − 2	1.77 E + 3	2.34 E + 9
Water	15.6	999	9.80	1.12 E − 3	1.12 E − 6	7.34 E − 2	1.77 E + 3	2.15 E + 9

^aIn contact with air.

^bIsentropic bulk modulus calculated from speed of sound.

^cTypical values. Properties of petroleum products vary.

Use these values for sanity checks, e.g. an answer for water that is far from 1000 kg/m³, or for mercury far from 13.6×10^3 kg/m³, is wrong before it is interpreted.

Properties of some common gases

■ **TABLE 1.8**
Approximate Physical Properties of Some Common Gases at Standard Atmospheric Pressure (SI Units)

Gas	Temperature (°C)	Density, ρ (kg/m ³)	Specific Weight, γ (N/m ³)	Dynamic Viscosity, μ (N · s/m ²)	Kinematic Viscosity, ν (m ² /s)	Gas Constant, ^a R (J/kg · K)	Specific Heat Ratio, ^b k
Air (standard)	15	1.23 E + 0	1.20 E + 1	1.79 E − 5	1.46 E − 5	2.869 E + 2	1.40
Carbon dioxide	20	1.83 E + 0	1.80 E + 1	1.47 E − 5	8.03 E − 6	1.889 E + 2	1.30
Helium	20	1.66 E − 1	1.63 E + 0	1.94 E − 5	1.15 E − 4	2.077 E + 3	1.66
Hydrogen	20	8.38 E − 2	8.22 E − 1	8.84 E − 6	1.05 E − 4	4.124 E + 3	1.41
Methane (natural gas)	20	6.67 E − 1	6.54 E + 0	1.10 E − 5	1.65 E − 5	5.183 E + 2	1.31
Nitrogen	20	1.16 E + 0	1.14 E + 1	1.76 E − 5	1.52 E − 5	2.968 E + 2	1.40
Oxygen	20	1.33 E + 0	1.30 E + 1	2.04 E − 5	1.53 E − 5	2.598 E + 2	1.40

^aValues of the gas constant are independent of temperature.
^bValues of the specific heat ratio depend only slightly on temperature.

The ideal gas law

Gases are highly compressible: their density changes directly with pressure and temperature through the equation of state

$$\rho = \frac{p}{RT}$$

with R the gas constant of the particular gas. The law is a good approximation for real gases away from liquefaction. In physics the same law is usually written per particle:

$$p = n k_B T$$

with n the number of molecules per unit volume, and with the Boltzmann constant in place of the gas constant.

$$k_B = 1.381 \times 10^{-23} \text{ J/K}, \quad N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

The two forms are the same equation

The number density is the mass density divided by the mass of one particle:

$$n = \frac{N}{V} = \frac{\rho}{m}, \quad m = \bar{\mu} m_u$$

with $\bar{\mu}$ the mean particle mass in atomic mass units and $m_u = 1.66054 \times 10^{-27}$ kg the atomic mass unit. For dry air $\bar{\mu} = 28.97$, so

$$m_{\text{air}} = (28.97)(1.66054 \times 10^{-27} \text{ kg}) = 4.81 \times 10^{-26} \text{ kg}$$

Substituting the number density into the second form:

$$p = n k_B T = \frac{\rho}{m} k_B T = \rho \underbrace{\frac{k_B}{m}}_R T \quad \Rightarrow \quad \rho = \frac{p}{RT}$$

So the gas constant of a particular gas is the Boltzmann constant per unit mass of particle, or the universal gas constant divided by the molar mass:

$$R = \frac{k_B}{m} = \frac{k_B}{\bar{\mu} m_u} = \frac{k_B N_A}{M} = \frac{R_u}{M} \quad R_{\text{air}} = \frac{8.314 \text{ J}/(\text{mol K})}{0.0290 \text{ kg/mol}} = 287 \text{ J}/(\text{kg K})$$

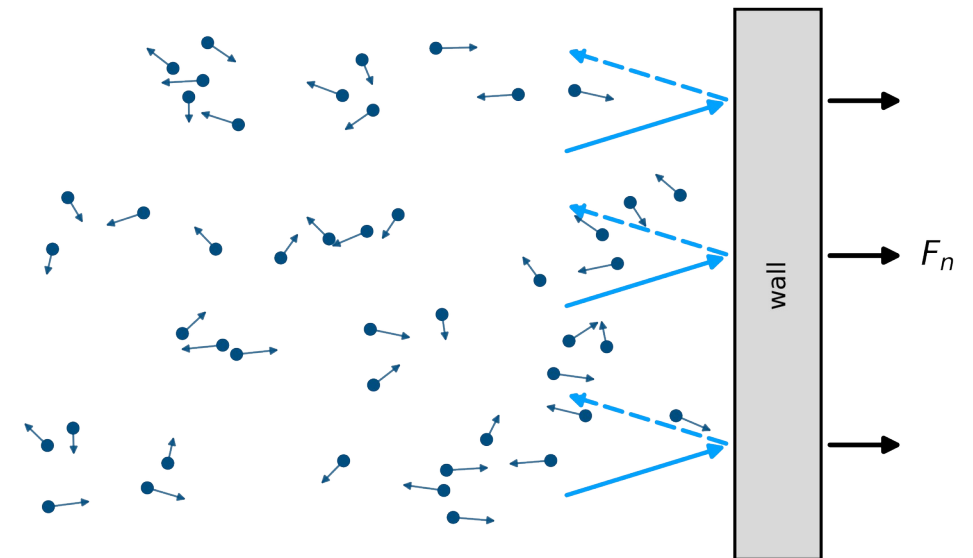
What is pressure?

A fluid exerts pressure because its molecules bombard the surfaces containing it. Each impact reverses the normal component of the molecule's momentum.

The impacts are far too frequent to feel one at a time: about 3×10^{23} molecules strike each square centimetre of wall per second, so the force is steady.

Pressure is that time-averaged normal force per unit area, acting along the normal to the surface whatever its orientation:

$$p = \frac{F_n}{A}$$



each impact reverses the normal momentum

SI units

$$1 \text{ Pa} = 1 \text{ N/m}^2, \quad 1 \text{ kPa} = 10^3 \text{ Pa}$$

British Gravitational units

$$1 \text{ psi} = 1 \text{ lb/in.}^2 = 6.895 \text{ kPa}$$

$$1 \text{ atm} = 101.325 \text{ kPa} = 14.696 \text{ psi}$$

Kinetic theory relates the bombardment to the temperature and recovers the gas law:

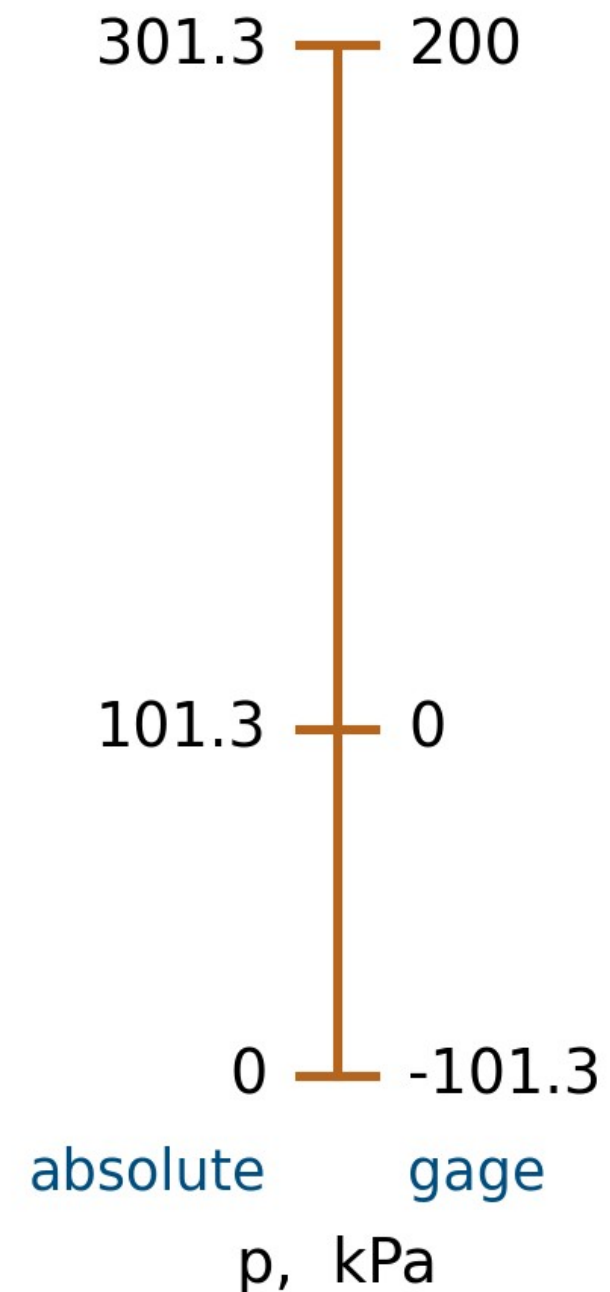
$$p = \frac{1}{3} n m \overline{v^2} = n k_B T \quad \text{because} \quad \frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T$$

Absolute and gage pressure

Both p and T in the gas law must be absolute: the pressure measured from a perfect vacuum, the temperature in kelvin.

$$p_{\text{abs}} = p_{\text{gage}} + p_{\text{atm}}$$

Gage pressure is measured from the local atmosphere, 101.3 kPa at sea level, so it is negative in a partial vacuum.



Viscosity



Viscous fluids.

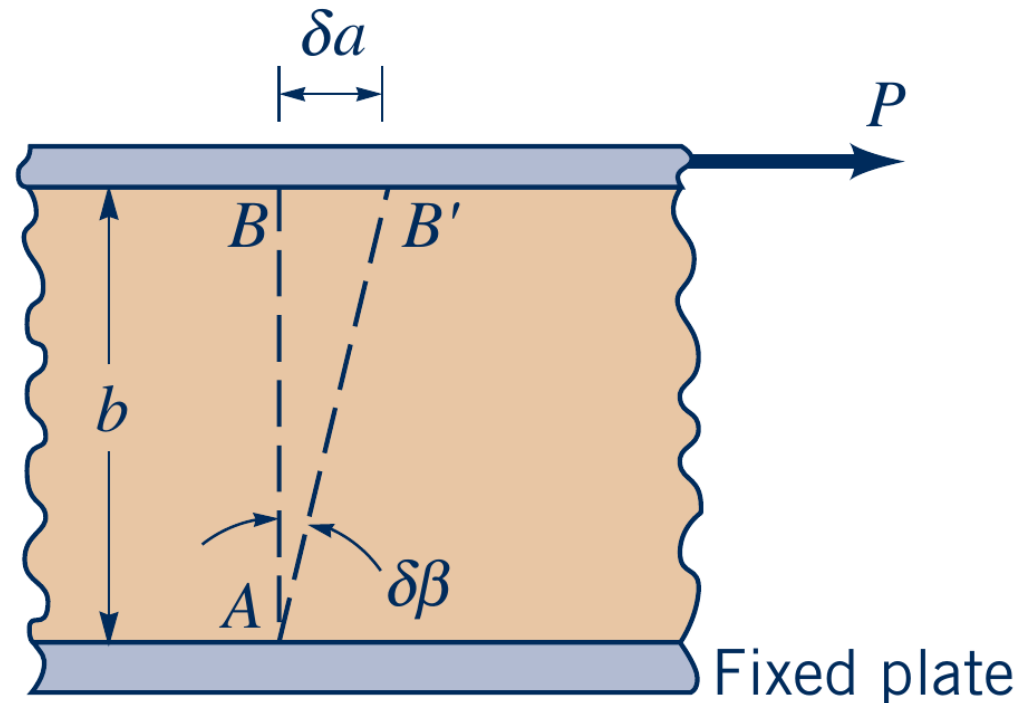
- The behaviour of a flowing fluid depends on various fluid properties.
- Viscosity, one of the important properties, is responsible for the shear force produced in a moving fluid.
- Although the two fluids shown look alike (both are clear liquids and have a specific gravity of 1), they behave very differently when set into motion.
- The very viscous silicone oil is approximately 10,000 times more viscous than the water.

Viscosity

Density and specific weight measure the heaviness of a fluid.

They are not enough: water and oil can have similar densities and behave quite differently when they flow.

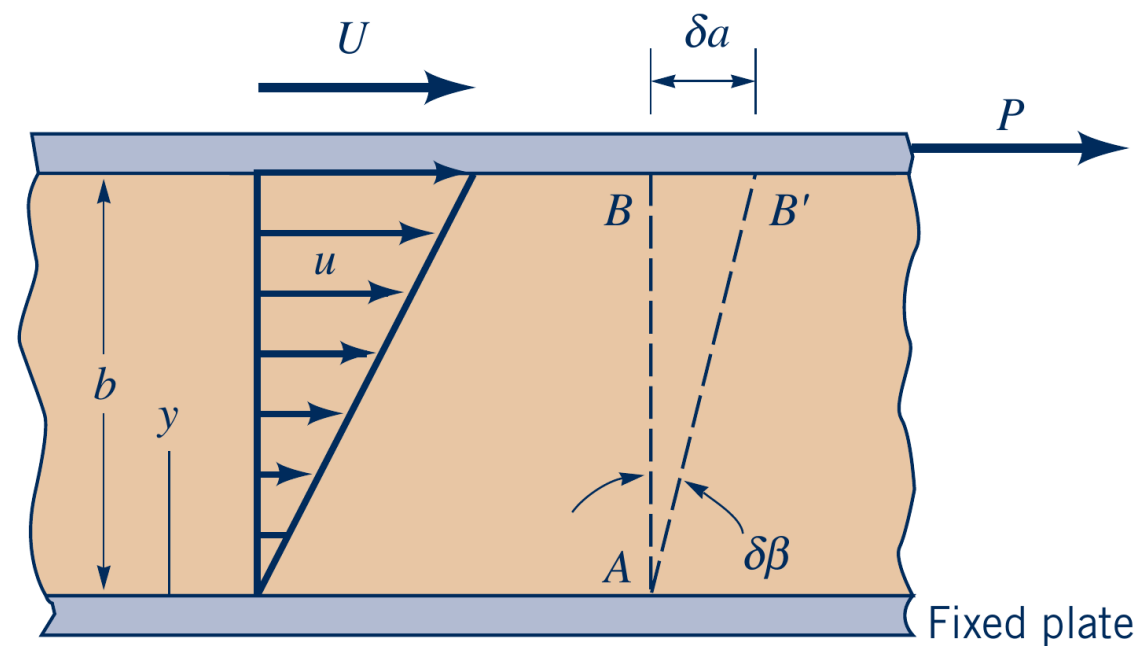
Place a material between two parallel plates, fix the lower one and pull the upper one with a force P . A solid deforms once, through a small angle $\delta\phi$



A solid placed between two parallel plates, and the forces on the upper plate.

A fluid between two plates

A fluid does not stop deforming: under the force P the upper plate keeps moving, with a constant velocity U .



The velocity varies across the gap, from the moving plate to the fixed one.

In a small time δt the line AB rotates through an angle $\delta\beta$:

$$\tan \delta\beta \approx \delta\beta = \frac{\delta a}{b} = \frac{U \delta t}{b}$$

The rate at which that angle changes is the rate of shearing strain:

$$\dot{\gamma} = \lim_{\delta t \rightarrow 0} \frac{\delta\beta}{\delta t} = \frac{U}{b} = \frac{du}{dy}$$

The no-slip condition



No-slip condition.

- As a fluid flows near a solid surface, it “sticks” to the surface: the fluid matches the velocity of the surface.
- This so-called no-slip condition is a very important one that must be satisfied in any accurate analysis of fluid flow phenomena.
- Dye injected at the bottom of a channel through which water is flowing forms a stagnant layer near the bottom, because of the no-slip condition.
- As the dye filament is moved away from the bottom, the motion of the water is clearly apparent. A significant velocity gradient is created near the bottom.

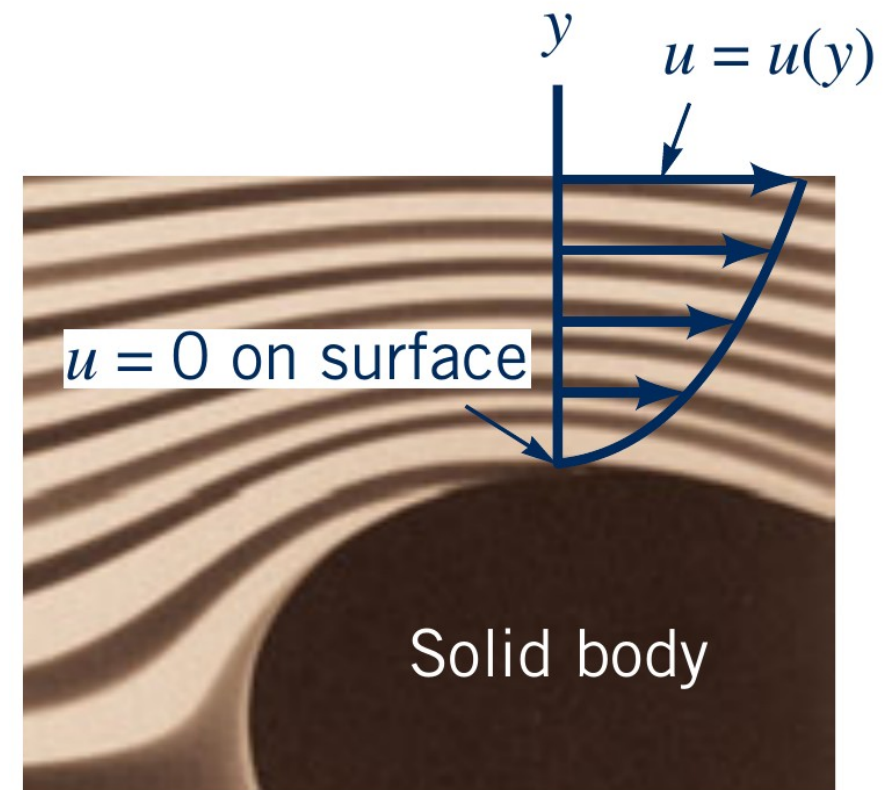
The no-slip condition

A fluid in contact with a solid boundary sticks to it: at the wall the fluid and the wall have the same velocity.

On a fixed wall the fluid velocity is zero. On the moving plate it equals the plate velocity U .

All fluids, liquids and gases alike, satisfy this condition. It is not an approximation.

Because of it a velocity gradient du/dy always develops next to a boundary, and with it a shearing stress. This is the boundary condition imposed on every wall in every simulation we will run.



Newtonian fluids

Increasing the force increases the rate of shearing strain in direct proportion:

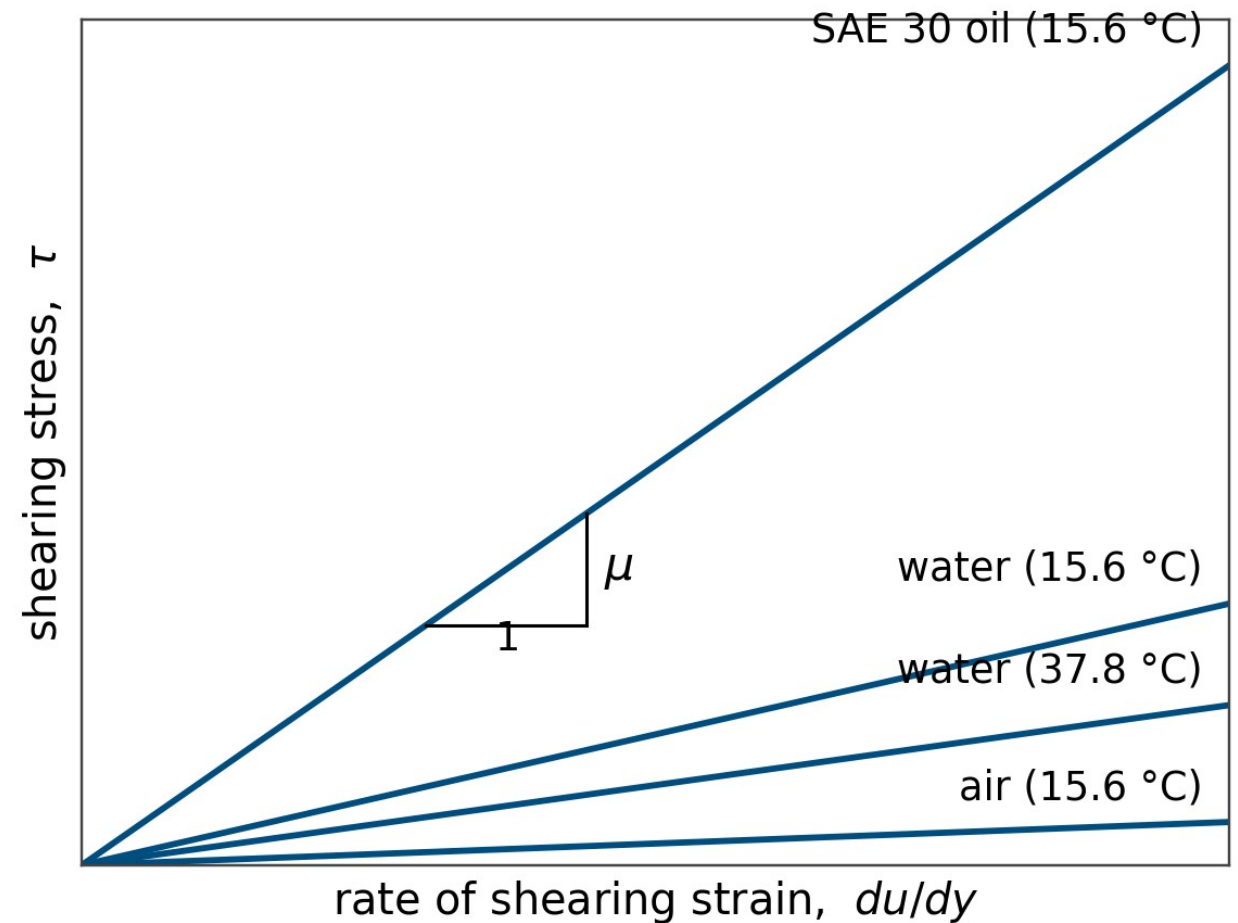
$$\tau \propto \dot{\gamma} = \frac{du}{dy}$$

For water, oil, gasoline and air the two are related by

$$\tau = \mu \frac{du}{dy}$$

μ is the absolute or dynamic viscosity.

Fluids that follow this law are Newtonian, and most common fluids are.



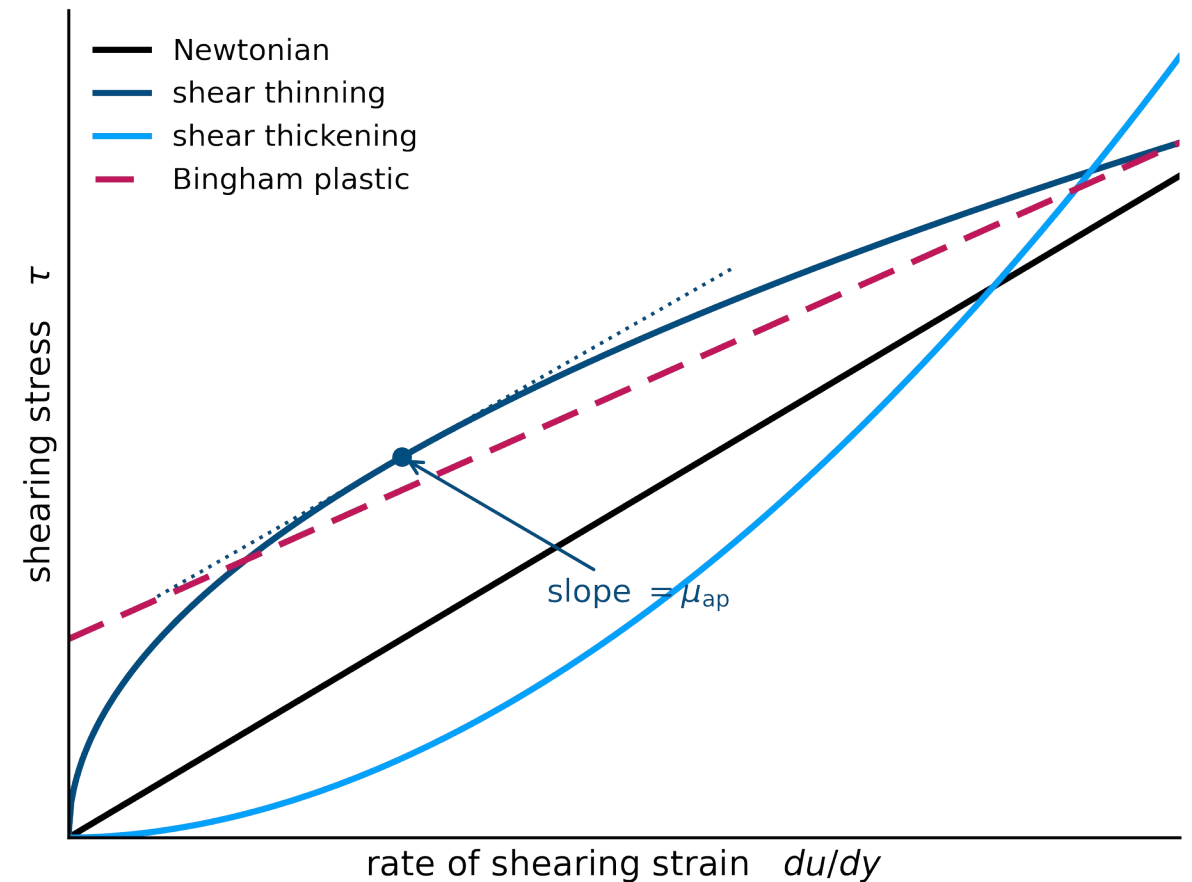
Linear variation of shearing stress with rate of shearing strain. The slope of each line is the viscosity.

Apparent viscosity

For any fluid, the ratio of the shearing stress to the rate of shearing strain is the apparent viscosity:

$$\mu_{ap} = \frac{\tau}{du/dy}$$

It is the slope of the curve of τ against du/dy at the point at which the fluid is being sheared, and it is what classifies the fluid.

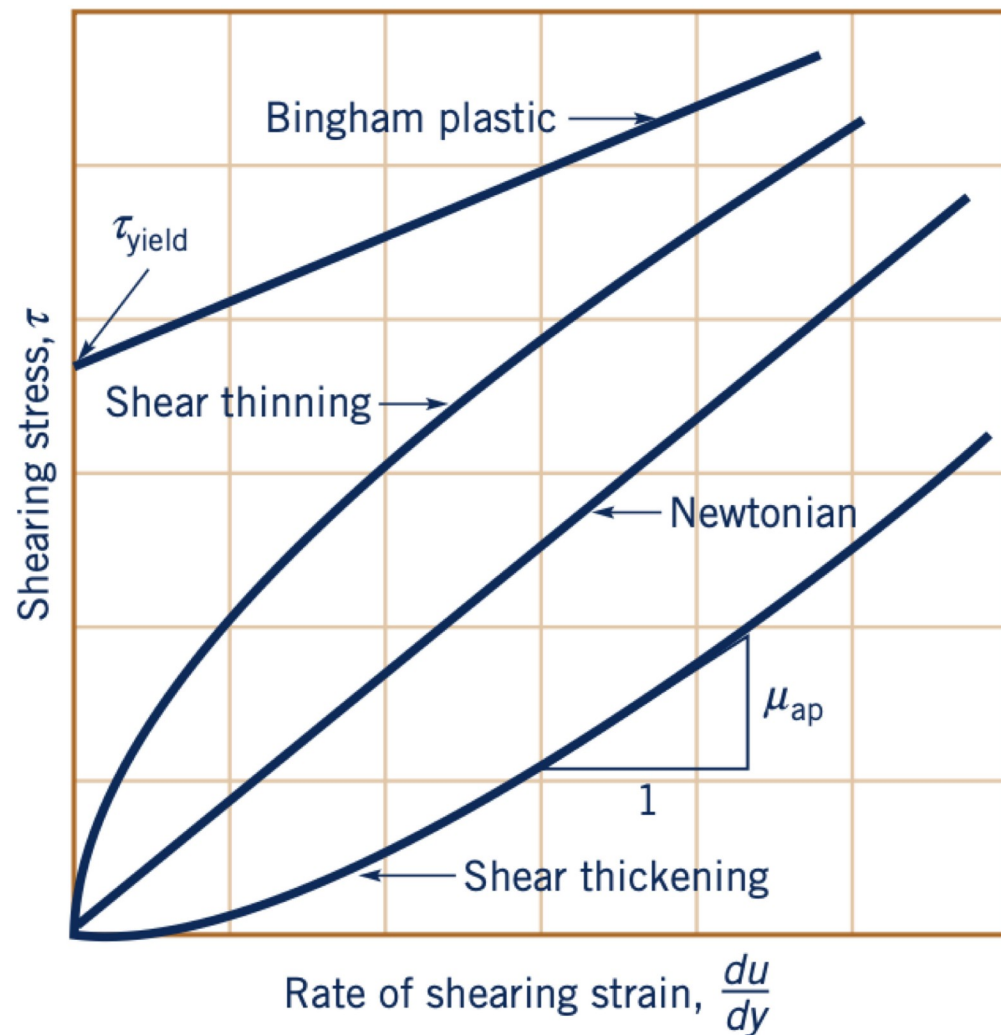


- **Constant slope:** the apparent viscosity does not depend on the shear rate, $\mu_{ap} = \mu$, and the fluid is Newtonian
- **Slope that falls with shear rate:** shear thinning, or pseudoplastic
- **Slope that rises with shear rate:** shear thickening, or dilatant
- **No flow at all below a yield stress:** Bingham plastic

For a non-Newtonian fluid a single number is not enough: the apparent viscosity must be quoted with the shear rate at which it was measured.

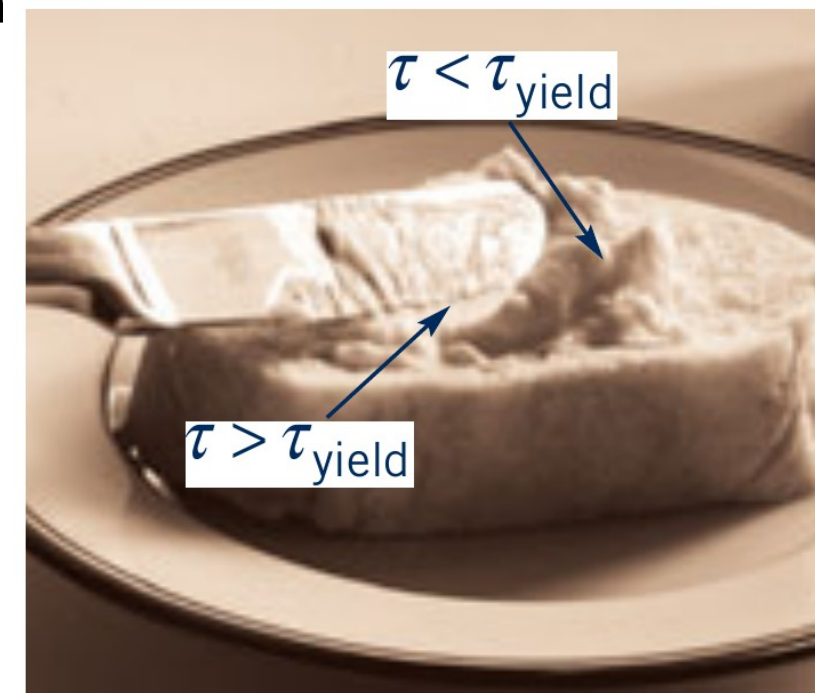
Non-Newtonian fluids

When the relation is not linear, the slope is called the **apparent viscosity** μ_{ap} and depends on the shear rate.



■ **Figure 1.7** Variation of shearing stress with rate of shearing strain for several types of fluids, including common non-Newtonian fluids.

- **Shear thinning:** the harder it is sheared the less viscous it becomes — **Examples:** latex paint, polymer solutions
- **Shear thickening:** the harder it is sheared the more viscous it becomes — **Examples:** corn starch in water, quicksand
- **Bingham plastic:** no motion below a yield stress, then it flows — **Examples:** toothpaste, mayonnaise



Non-Newtonian behaviour



Non-Newtonian behaviour.

- Fluids for which the shearing stress is not linearly related to the rate of shearing strain are designated as **non-Newtonian fluids**.
- A mixture of water and corn starch, when placed on a flat surface, flows as a thick, viscous fluid.
- When it is rapidly disturbed, it appears to fracture and behave more like a solid.
- **The mixture is a shear thickening fluid:** it becomes more viscous as the shearing rate is suddenly increased through the rapid action of the spoon.

Viscosity and temperature

Dimensions and units of the dynamic viscosity:

$$[\mu] \doteq FTL^{-2} \doteq ML^{-1}T^{-1} \quad : \quad \text{Pa} \cdot \text{s} = \text{N} \cdot \text{s}/\text{m}^2 = \text{kg}/(\text{m s})$$

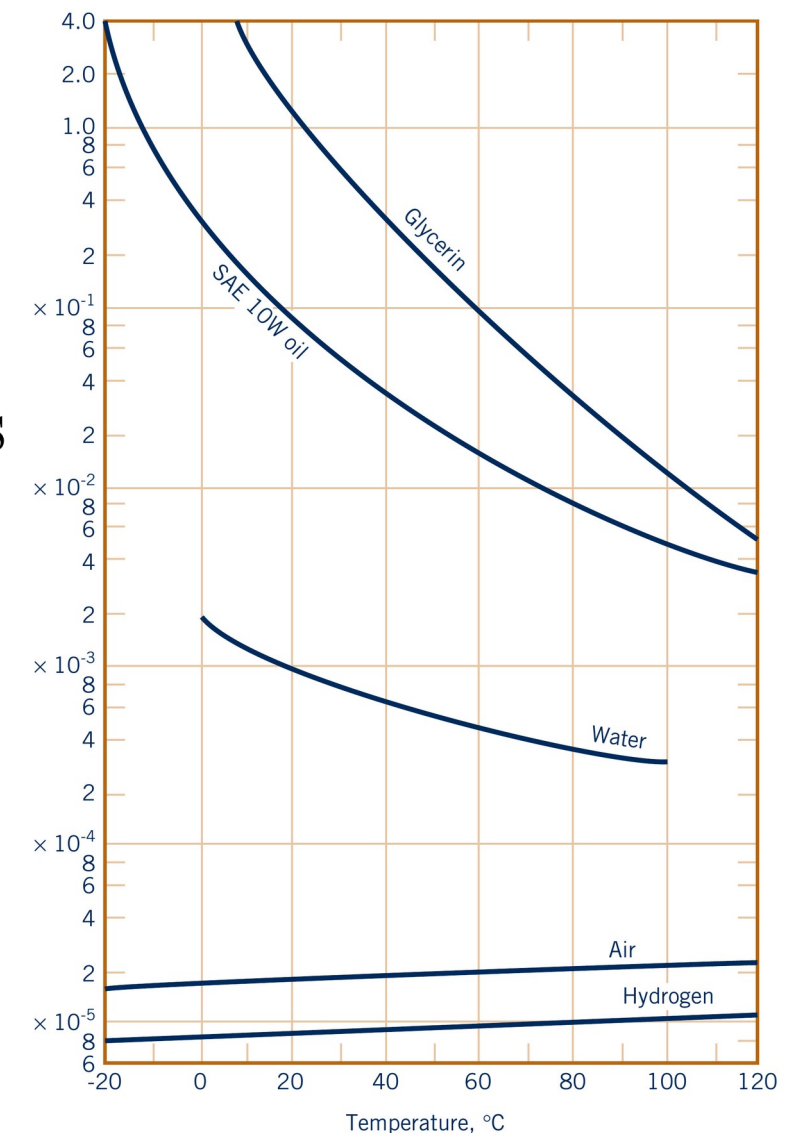
In the CGS system the unit is the poise, still met in the older literature:

$$1 \text{ P} = 1 \text{ poise} = 0.1 \text{ Pa} \cdot \text{s}, \quad 1 \text{ cP} = 10^{-3} \text{ Pa} \cdot \text{s}$$

Viscosity depends only mildly on pressure, and that effect is usually neglected.

It is very sensitive to temperature:

between 15.6 °C and 37.8 °C the density of water falls by less than 1%, but its viscosity falls by about 40%.



$$\mu_{\text{H}_2\text{O}} = 1.00 \times 10^{-3} \text{ Pa} \cdot \text{s}, \quad \mu_{\text{air}} = 1.82 \times 10^{-5} \text{ Pa} \cdot \text{s} \quad (20^\circ\text{C})$$

The curves for liquids and gases run in opposite directions.

Temperature: liquids and gases

The difference in the effect of temperature on the viscosity of liquids and gases can be traced back to the difference in molecular structure.

→ LIQUIDS

- The molecules are closely spaced, with strong cohesive forces between them.
- The resistance to shearing comes from those cohesive forces.
- Raising the temperature weakens them, so the viscosity falls.

Andrade's equation, with D and B constants of the liquid:

$$\mu = D e^{B/T}$$

→ GASES

- The molecules are widely spaced, and the cohesive forces are negligible.
- The resistance to shearing comes from the momentum exchanged between layers by molecules crossing from one to the other.
- Raising the temperature increases the random motion, so the exchange grows and the viscosity rises.

Sutherland's equation, with C and S constants of the gas:

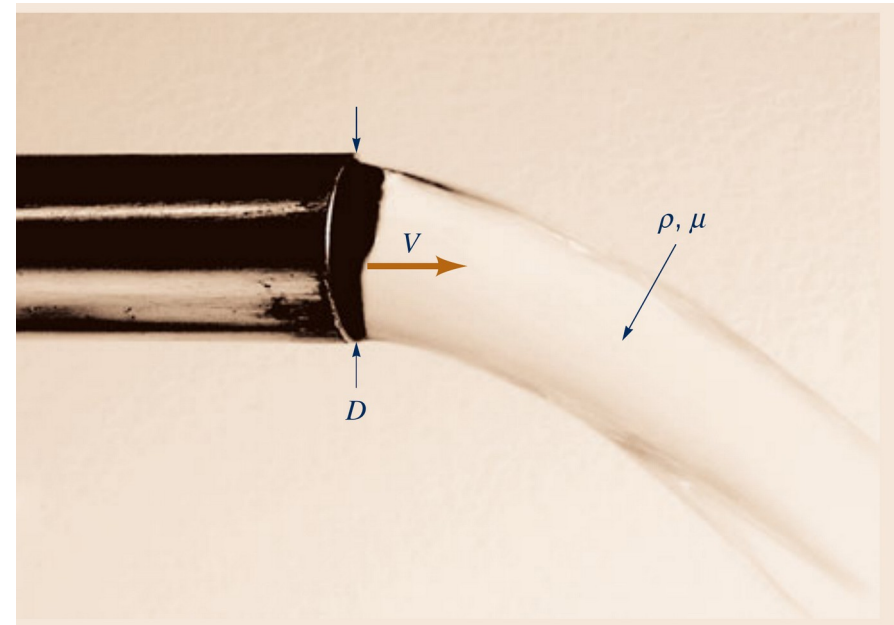
$$\mu = \frac{C T^{3/2}}{T + S}$$

Kinematic viscosity and the Reynolds number

Viscosity almost always appears divided by the density:

$$\nu = \frac{\mu}{\rho}$$

ν is the kinematic viscosity, with dimensions L^2T^{-1} , measured in m^2/s .



It is the combination that appears in the most important dimensionless group of the course, the Reynolds number:

$$Re = \frac{\rho v D}{\mu}$$

It compares inertia with viscous friction, and it decides whether a flow is laminar or turbulent.

Example — Newtonian fluid shear stress

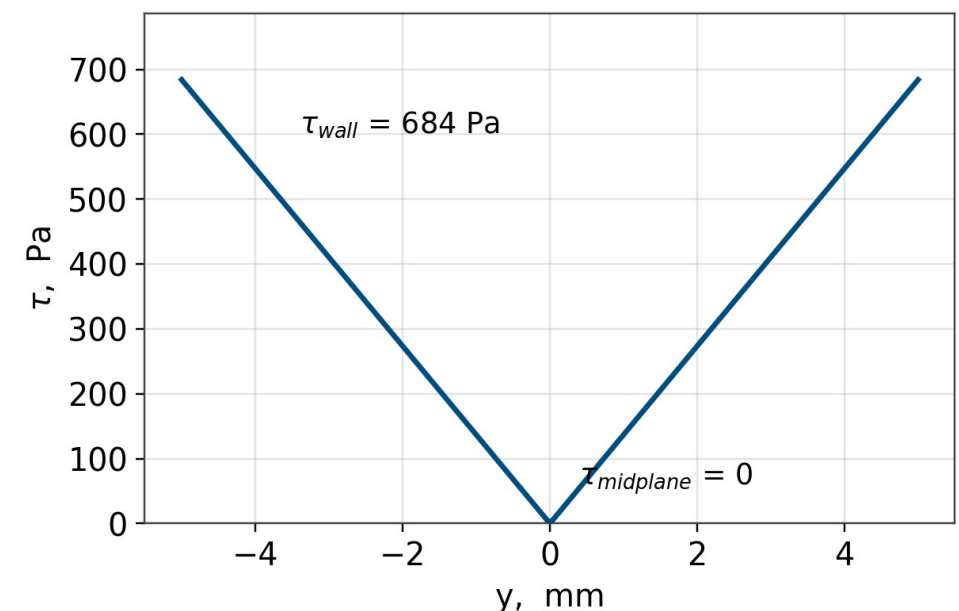
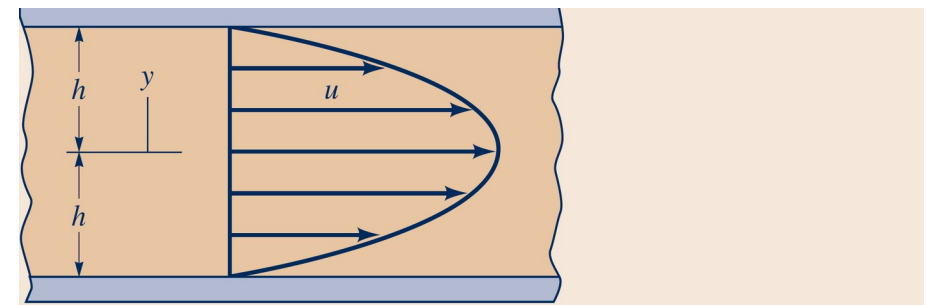
The velocity distribution for the laminar flow of a Newtonian fluid between two fixed wide parallel plates is $u = (3v/2)[1 - (y/h)^2]$, where v is the mean velocity, h the half-distance between the plates and y is measured from the centreline. The fluid has $\mu = 1.9 \text{ Pa}\cdot\text{s}$, with $v = 0.60 \text{ m/s}$ and $h = 5.0 \text{ mm}$. Determine the shearing stress at the wall and at the midplane.

$$u = \frac{3v}{2} \left[1 - \left(\frac{y}{h} \right)^2 \right]$$

$$\frac{du}{dy} = -\frac{3vy}{h^2}$$

$$\tau_{\text{wall}} = \mu \frac{3v}{h} = \frac{(1.9)(3)(0.60)}{5.0 \times 10^{-3}} = 684 \text{ Pa}$$

$$\tau_{\text{midplane}} = 0$$



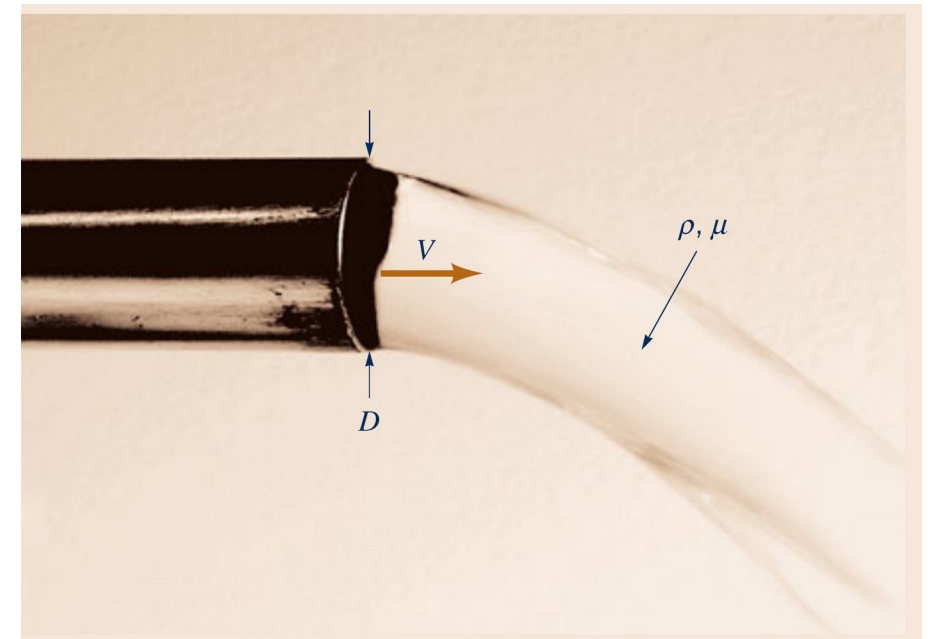
Example — the Reynolds number

A Newtonian fluid with a dynamic viscosity of $0.38 \text{ N}\cdot\text{s}/\text{m}^2$ and a specific gravity of 0.91 flows through a 25 mm diameter pipe with a mean velocity of 2.6 m/s. Determine the value of the Reynolds number.

The density follows from the specific gravity:

$$\rho = SG \rho_{\text{H}_2\text{O}} = 0.91 (1000) = 910 \text{ kg}/\text{m}^3$$

$$\text{Re} = \frac{(910 \text{ kg}/\text{m}^3)(2.6 \text{ m}/\text{s})(0.025 \text{ m})}{0.38 \text{ N}\cdot\text{s}/\text{m}^2} = 156$$



The Reynolds number is dimensionless: the same value is obtained in any consistent system of units, provided every variable is converted first.

Experimental determination of viscosity



Capillary tube viscometer.

- Most devices used to determine viscosity, called viscometers, do not measure it directly: **they measure some characteristic with a known relationship to viscosity.**
- The capillary tube viscometer involves the laminar flow of a fixed volume of fluid through a capillary tube.
- The time required for the fluid to pass through the tube is a measure of the kinematic viscosity of the fluid.
- As shown with the four tubes, the drain times vary with the viscosity of the fluid and the diameter of the capillary tube.

Lecture summary

A fluid deforms continuously under a shearing stress of any magnitude. Its behaviour is described by properties defined as averages over a continuum, in equations that are dimensionally homogeneous and solved in one consistent system of units.

$$\rho = \frac{m}{V}$$

density

$$p = \frac{F_n}{A}$$

pressure

$$p = n k_B T$$

gas law per particle

$$\nu = \frac{\mu}{\rho}$$

kinematic viscosity

$$\gamma = \rho g$$

specific weight

$$p_{\text{abs}} = p_{\text{gage}} + p_{\text{atm}}$$

absolute pressure

$$\tau = \mu \frac{du}{dy}$$

Newtonian shear stress

$$SG = \frac{\rho}{\rho_{\text{H}_2\text{O}}}$$

specific gravity

$$\rho = \frac{p}{RT}$$

ideal gas law

$$\mu_{\text{ap}} = \frac{\tau}{du/dy}$$

apparent viscosity

$$\text{Re} = \frac{\rho v D}{\mu}$$

Reynolds number